

SCI-FLT-FIXED v0.1 Engineering-Conformance Specification

SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/draft-r0.2

Status: implementation-blind Stage B conformance-target draft; no conformity finding

Scientific owner: Grant Wilson

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Source SHA-256: 1c26c9986b2ef06f9cd68d21b2454833b8d092d73a167b32cbb89bf4f075b73c
Shared normative core SHA-256: bbcce3546d895002af8f7bb847f32e734cb23e6b4963132a09e6f54ffe4d6481
Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
Owner directive SHA-256: 02ca63dcc8ac19e554fa333c20773ff85264336b4f622024605f6514ec1716d1
Build recipe SHA-256: 97b897beb2638d2b2081737fa8144a7e63d95f13eabde8f3bba774dc38a7cef5

Implementation-blind scientific-contract draft. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, performance, readiness, freeze, production, or Unity result.

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Normative import: the complete SCI-FLT-FIXED-NORMATIVE-CORE v0.1/draft-r0.2, source SHA-256 bbcce3546d895002af8f7bb847f32e734cb23e6b4963132a09e6f54ffe4d6481, is incorporated without modification. This view adds no scientific rule. If this view and the imported core differ, the core controls.

1. Conformance target and evidence boundary

This specification translates the shared scientific core into observable engineering obligations without describing, inspecting, or judging an implementation. A future conformance review must bind one candidate realization to one exact core revision, parent generation, resolved operator generation, and product generation.

A claim passes only when positive evidence establishes the exact required identity and behavior. A missing, conflicting, approximate, same-name, or inferred fact yields unavailable or failed status as directed by the core. A finite array, successful process exit, or familiar label is not sufficient.

Conformance, scientific validation, calibration, performance, readiness, and production are separate evidence classes. This specification defines only a future conformance target.

2. Required input and plan records

A candidate realization must accept an immutable request record, effective decision, complete parent binding, and externally resolved fixed plan. The records must expose enough content to reconstruct and compare exact identities, not merely display friendly names.

The parent record must distinguish MAP observation, MAP coadd, and JINC observation roles. It must bind the complete applicable upstream identity, quantity, units, nominal-beam, WCS, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance state. JINC requires all five atomic numerical roles. Upstream unavailability must remain explicit.

The plan record must distinguish requested, effective, disabled, unavailable, and resolved states. A resolved plan must identify FLT-FIXED-CONV as the sole base family, bind one exact sampled convolution, and bind every coefficient, parameter, coordinate domain, grid and metric fact, normalization, support and edge rule, transfer qualification, and provenance fact before application. "Externally resolved" does not transfer FLT policy or application ownership.

3. Operator reconstruction record

FLT-OPERATOR must provide a content-bound representation sufficient to reconstruct the exact finite L_{Theta} , frozen J_{full} , and complete A_{Theta}, J , including:

- ordered input and output row domains;
- exact WCS, frame, topology, metric, shape, indexing, and pixel-area facts;
- complete sampled coefficients, coefficient units and ordering, numerical representation, and digest;
- $K_{\text{geom}}, K_{\text{nonzero}}, K_{\text{req}}$, their relations, exact-zero representation, center, extent, tie, phase, subpixel convention, orientation, and handedness;
- declared normalization, DC gain, and distinct signed, absolute, squared, and geometric support summaries;
- full-footprint predicates, exact inherited zero-operator row domain, and cause vocabulary;
- qualified transfer facts or explicit unavailability; and
- immutable plan and operator generation identifiers.

The record must show that one product applies this exact sampled convolution once and makes no intermediate or reordered-composition claim. An alternate computational mechanism is comparable only after reconstructing the identical declared operator. A different coefficient representation that changes a declared coefficient, ordering, phase, row domain, or numerical operator is not scientific equivalence.

4. Admission and application gate

Bundle admission must evaluate SCI-FLT-FIXED:input_bundle_admission@1 against one exact request, parent, and resolved plan. Row admission must evaluate SCI-FLT-FIXED:input_row_admission@1 for every candidate row against the frozen plan and exact K_req. Each returns the typed request, applicability, eligibility, missing/conflict, action, and cause state defined by its bound profile. Neither performs payload arithmetic.

Application may proceed only for an eligible resolved route. The selector must be resolved once from immutable membership before payload arithmetic. For every output row, it verifies all K_req locations against the exact parent domain, FLT input admission, availability, finiteness, and required predicates. The scientific row domain must equal the passing set. Response perturbations, covariance draws, noise realizations, and NOI members must reuse that selector; a member failure produces row unavailability, not re-selection.

No extension, periodic wrap, truncation, local renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is conforming in v0.1. Rows failing the full-footprint gate remain unavailable with causes, even if a storage payload is finite.

5. Signal, response, transfer, mode, and influence records

Application must compute FLT-SIG from the exact resolved operator on the exact scientific row ordering. The output-unit record must derive units from parent and coefficient units and retain the originating nominal-beam and calibration lineage without introducing a filtered-beam label.

Each response record must name fixed-state linear, already realized parent- grid, parent full-procedure finite difference with FLT fixed, or FLT re- resolved procedure family. The first three must apply the identical frozen A_Theta, J exactly once; the fourth is outside v0.1. Missing basis, domain, or parent-response identity is explicitly unavailable. The exact zero operator records its local zero derivative without promoting it to an unavailable complete source response. Kernel-only, approximate, multiply applied, differently centered, differently normalized, or differently edged surrogates do not pass.

Transfer and mode records must bind the exact coordinate domain, WCS metric, frequency grid, finite-grid domain, phase, normalization, null, invariant, anisotropy, and attenuation facts that are scientifically defined. They must distinguish sampled-kernel transfer from the complete row-restricted operator. Unsupported facts remain unavailable. Influence exposes the parent-row coefficient relation; FLT-EXPOSURE-LINEAGE binds parent exposure identity or typed absence and confirms that neither influence nor a convolved exposure plane is physical exposure.

6. Covariance and NOI records

For an available compatible parent covariance, a covariance-qualified candidate must realize the exact two-sided frozen-operator propagation on S_out. Its record must state parent stochastic authority separately from output representation and name the conditional model, omitted terms, excluded selection/kernel/beam/WCS/calibration/model uncertainties, domain, ordering, rank, null space, and supported operations.

If an explicitly diagonal parent covariance is propagated, any complete output representation must include induced cross-row covariance. A marginal-only representation must say so. Missing cross terms may not be encoded as zero or independence.

FLT-NOI-ATTACHMENT-STATE must be an FLT relation record, not the separately owned NOI product and not an FLT completion dependency. Every admitted NOI member must receive the identical frozen A_Theta, J;

member footprint failure is unavailable. The route rejects filtering uncertainty-like products, approximate transfer, relocation, commutation, substitution, and any per-member re-resolution.

7. Atomic publication and lifecycle records

Publication must evaluate SCI-FLT-FIXED:output_publication@1 against one complete publication candidate. The candidate contains all required roles from the imported core, including request-specific honest unavailable companion records. A partial candidate, placeholder, or inferred companion fails publication; a successful publication action creates the realized product.

Lifecycle evidence must distinguish not_requested, requested, effective, disabled, unavailable, resolved, applied, complete_publication_candidate, publication_decision, not_produced, realization_failed, failed, realized_identity, realized_zero, realized, and superseded. Disabled is not produced. Identity and zero use the same candidate and decision sequence. Required failure propagates and emits no complete product.

Every product must bind immutable parent, plan, operator, output, companion, cause, lifecycle, failure, and provenance generations. Any core-defined identity change creates a new generation. A later NOI attachment is a separate companion and cannot mutate an existing FLT bundle.

8. Requirement conformance matrix

The following matrix routes every stable core requirement to a future observable conformance decision. The exact normative text remains in the imported core.

- SCI-FLT-FIXED-REQ-001: verify package, tranche, convolution-only base, and qualified low-pass subtype identity; reject arbitrary-linear or inference- bearing identity.
- SCI-FLT-FIXED-REQ-002: reconstruct A and verify no additive term or additive-state dependency.
- SCI-FLT-FIXED-REQ-003: verify exactly one supported immutable parent role.
- SCI-FLT-FIXED-REQ-004: verify complete applicable parent fields and atomic JINC roles.
- SCI-FLT-FIXED-REQ-005: verify upstream unavailability remains fail-closed.
- SCI-FLT-FIXED-REQ-006: verify observation, coadd, and JINC successor identities and absence of FLT coaddition or inferred commutation.
- SCI-FLT-FIXED-REQ-007: verify all plan and selector state predates payload arithmetic and is independent of response, covariance, noise, and NOI member payloads.
- SCI-FLT-FIXED-REQ-008: compare exact WCS and grid facts and reject approximate joins or resampling.
- SCI-FLT-FIXED-REQ-009: reconstruct exact finite coefficients, coordinate domain, three support sets, domains, discretization, and digest.
- SCI-FLT-FIXED-REQ-010: verify one-time convolution construction from the exact sampled kernel without intermediate or reordered claims.
- SCI-FLT-FIXED-REQ-011: verify every low-pass transfer fact or mark only the qualification unavailable.
- SCI-FLT-FIXED-REQ-012: recompute the full-footprint row set from exact K_req and predicates, including exact-zero stored coefficients.
- SCI-FLT-FIXED-REQ-013: verify all excluded rows are unavailable with causes, never promoted by storage shape.
- SCI-FLT-FIXED-REQ-014: detect and reject every deferred edge or replacement method.
- SCI-FLT-FIXED-REQ-015: verify transformed-amplitude identity and exact unit derivation without amplitude-category relabeling.

- SCI-FLT-FIXED-REQ-016: verify nominal-beam and CAL lineage retention and absence of unsupported beam or flux claims.
- SCI-FLT-FIXED-REQ-017: identify the response family, compare its one-time frozen-operator composition, or verify honest unavailability.
- SCI-FLT-FIXED-REQ-018: verify finite-domain transfer, mode, and phase facts or explicit unavailable states without whole-chain promotion.
- SCI-FLT-FIXED-REQ-019: compare influence to coefficients, verify it is not exposure, and verify exact exposure-lineage state.
- SCI-FLT-FIXED-REQ-020: verify distinct computability, support, admission, validity, confidence, and downstream states.
- SCI-FLT-FIXED-REQ-021: compare covariance to exact two-sided propagation.
- SCI-FLT-FIXED-REQ-022: verify separate parent-authority and output- representation axes, conditional model, exclusions, and unknown cross-term handling.
- SCI-FLT-FIXED-REQ-023: test induced off-diagonal terms and reject marginal planes presented as complete covariance.
- SCI-FLT-FIXED-REQ-024: verify empirical uncertainty and significance are absent from FLT ownership.
- SCI-FLT-FIXED-REQ-025: compare every NOI member's full operator state to the signal state.
- SCI-FLT-FIXED-REQ-026: detect and reject per-member selection or re-resolution.
- SCI-FLT-FIXED-REQ-027: verify every atomic role, exposure lineage, allowed honest unavailable record, and NOI attachment-state relation without an NOI completion dependency.
- SCI-FLT-FIXED-REQ-028: verify the complete candidate-before-publication lifecycle vocabulary, causes, failure, and generation bindings.
- SCI-FLT-FIXED-REQ-029: test distinct disabled, identity, and zero outcomes.
- SCI-FLT-FIXED-REQ-030: mutate each identity-defining fact in isolation and verify a new immutable generation; verify NOI nonmutation.
- SCI-FLT-FIXED-REQ-031: remove or conflict each required fact and verify fail-closed behavior without fallback.
- SCI-FLT-FIXED-REQ-032: evaluate distinct bundle and row admission request, applicability, eligibility, unavailable, exclusion, and cause branches.
- SCI-FLT-FIXED-REQ-033: evaluate the complete publication candidate, request-specific honest-unavailable, partial, disabled, identity, zero, and failed publication branches.
- SCI-FLT-FIXED-REQ-034: verify VAL only binds or evaluates an immutable approved policy and authors no FLT fact or arithmetic.
- SCI-FLT-FIXED-REQ-035: verify no generic downstream admission follows from product availability.
- SCI-FLT-FIXED-REQ-036: verify every excluded family and every Stage B nonclaim remains outside the product claim set.
- SCI-FLT-FIXED-REQ-037: freeze J_full once and verify identical selector reuse plus explicit exclusion of unsupplied selection uncertainty.
- SCI-FLT-FIXED-REQ-038: verify K_geom, K_nonzero, K_req, exact-zero, identity, and declared zero-operator support semantics.
- SCI-FLT-FIXED-REQ-039: evaluate all three typed policy objects and each request-specific companion qualification.

- SCI-FLT-FIXED-REQ-040: distinguish every response family and prevent local zero derivative from becoming an unsupported complete response claim.
- SCI-FLT-FIXED-REQ-041: verify exactly one sampled convolution application and absence of composition, collapse, or reordering claims.
- SCI-FLT-FIXED-REQ-042: verify preregistered numerical comparison fields, frozen bounds, independent oracle, and nonclaim boundary.
- SCI-FLT-FIXED-REQ-043: verify exact inherited exposure lineage and reject influence or convolved exposure as physical exposure.
- SCI-FLT-FIXED-REQ-044: verify external timing while retaining exact FLT-owned plan, selector, application, and publication ownership.

9. Falsifiable prediction suite

Each future prediction evaluation must bind exact inputs, operator state, expected row domain, expected companions, expected lifecycle, comparison semantics, and causes before execution. The normative expected outcomes are in the imported core.

- SCI-FLT-FIXED-PRED-001: identity operator and identity lifecycle.
- SCI-FLT-FIXED-PRED-002: zero operator, inherited parent-support domain, local zero derivative and covariance, honest unavailable complete response, and zero lifecycle.
- SCI-FLT-FIXED-PRED-003: scalar input linearity on one frozen selector.
- SCI-FLT-FIXED-PRED-004: constant input under exact DC gain and every authorized normalization.
- SCI-FLT-FIXED-PRED-005: impulse, center, orientation, phase, and indexing.
- SCI-FLT-FIXED-PRED-006: exact separately typed response-family composition with one frozen operator application.
- SCI-FLT-FIXED-PRED-007: signed coefficients and distinct support summaries.
- SCI-FLT-FIXED-PRED-008: zero-sum constant-mode null without claim promotion.
- SCI-FLT-FIXED-PRED-009: full-footprint inclusion and dependent-row removal.
- SCI-FLT-FIXED-PRED-010: rejection of every deferred edge and replacement method, including support-conditioned renormalization.
- SCI-FLT-FIXED-PRED-011: missing, unavailable, non-admitted, and non-finite required parent locations.
- SCI-FLT-FIXED-PRED-012: exact complete covariance on ordered S_{out} .
- SCI-FLT-FIXED-PRED-013: induced cross-row covariance conditional on an explicit independent-diagonal parent model.
- SCI-FLT-FIXED-PRED-014: unavailable parent response or covariance and rejection of surrogates.
- SCI-FLT-FIXED-PRED-015: exact WCS and grid mismatch failure.
- SCI-FLT-FIXED-PRED-016: distinct observation, coadd, and JINC identities with no FLT coadd product.
- SCI-FLT-FIXED-PRED-017: exact frozen-selector NOI member parity and member-level unavailable footprint state.
- SCI-FLT-FIXED-PRED-018: per-member re-resolution rejection.
- SCI-FLT-FIXED-PRED-019: disabled, identity, zero, and failure lifecycle distinctions.
- SCI-FLT-FIXED-PRED-020: upstream unavailable MAP and JINC route.
- SCI-FLT-FIXED-PRED-021: low-pass qualification completeness.

- SCI-FLT-FIXED-PRED-022: exact-zero stored-coefficient required dependency.
- SCI-FLT-FIXED-PRED-023: zero-operator inherited row-support domain.
- SCI-FLT-FIXED-PRED-024: independent exact sampled low-pass transfer on the declared frequency grid and metric.

10. Numerical-comparison evidence record

A future candidate comparison must bind one immutable policy before results are observed. The evidence record must identify its independent exact or high- precision oracle; absolute, relative, and near-zero rules; signed-cancellation and zero-sum cases; conditioning and operation-count dependence; covariance comparison; sequential and parallel agreement; overflow, underflow, and non- finite handling; simultaneous row decisions; lifecycle; and provenance.

The record must prove that bounds were not changed after a failure and that scientific coefficients and identity remained exact. Passing such a policy would be future conformance evidence only. This specification supplies no candidate result, numerical-adequacy finding, validation, or performance claim.

11. Traceability, identity, and reproducibility record

TRACEABILITY.json maps every requirement and prediction identifier to its normative-core section, scientist-rationale section, engineering-conformance section, exact admitted Stage A object sections, and r0.2 owner-directive sections where the revision changes semantics. The durable verifier must reject missing, duplicate, extra, malformed, or unadmitted trace entries.

BUILD_BINDING.json records the packet manifest binding, all 17 admitted object hashes and byte counts, exact r0.2 owner-directive binding, every Stage B source and build-tool hash, deterministic build environment, output PDF identities, page counts, byte sizes, and SHA-256 digests. The source-closure, owner-decision parity, core/view parity, semantic-change, profile, rebuild, and visual-QA records are required. A clean separate rebuild must reproduce the three PDF digests exactly.

PDF text must expose the document identity, exact source digest, exact shared core digest, exact packet-manifest digest, and exact builder digest. PDF page rendering and visual QA are required in addition to content and hash checks.

12. Nonclaims

This conformance specification does not identify, inspect, modify, or judge an implementation. It records no implementation conformity, algorithm change, validation result, calibration state, achieved response or covariance, numerical adequacy, performance, readiness, scientific freeze, production status, or Unity activity.